

Portfolio

Conception Phase for Habit Tracking App

Course: DLBDSOOFPP01 – Object Oriented and Functional Programming with Python

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Overview

The Habit Tracking App is intended to help users create, monitor, and analyze their habits. It lets users set up habits with specific frequencies and log when they complete them, providing insights on how habits are formed and maintained. This conception phase overview covers the basic technical setup for developing the app, including the main class structures, data storage methods, user interactions, and overall workflow.

Core Components

1. Habit Class

The Habit class contains all necessary attributes (data) and behaviors (methods) for a habit. Using a class, such as the Habit class, is important in object-oriented programming because it combines related data and functions which make the code more organized and reusable.

- **Attributes:**
 - The name of the habit.
 - The date when the habit tracking will start.
 - The frequency at which the habit should be completed (daily or weekly).
 - A list of dates and times when the habit will be completed or logged.
- **Methods:**
 - Mark the habit as completed on a given date.
 - Calculate the current streak of the habit.
 - Determine if the habit was broken based on the periodicity and log.

2. Data Storage and Retrieval

The habit data needs to be stored between user sessions. Using a JSON file simplifies the implementation of habit data storage, structuring the data as a list of habit dictionaries.

3. User Interaction

The main.py module of the habit tracking app will provide a command-line interface (CLI) through the questionnaire library that prompts users for their input. Its functionalities include adding habits, viewing habits, logging entries, analyzing habits, and deleting habits. Error handling will be implemented to ensure that if the user makes a mistake, they will return to the main menu.

- **CLI Commands:**

- Add a new habit.
- View all habits with details such as name, start date, and periodicity.
- Log a completion date and time for a habit.
- Analyze habits to get insights like the longest run streak.
- Remove a habit from the system.
- Exit the application.

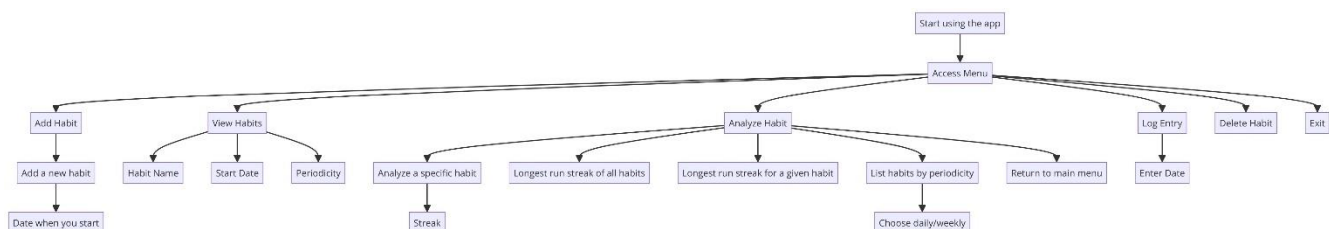
4. Analytics Module

The analytics module is for providing various functionalities to analyze the habits tracked by the app. It gives users an overview of their habit performance and consistency. Specifically, it can:

- Return a list of all currently tracked habits.
- Filter and return habits based on their periodicity (daily or weekly).
- Identify the habit with the longest streak.
- Calculate the longest streak for a specified habit.

Flow Chart Diagram

The following flow chart diagram illustrates the interaction between the user and the application:



- **Start using the app:** The user begins by accessing the main menu.
- **Access Menu:** From the main menu, the user can choose various options:
 - **Add Habit:** Allows the user to add a new habit, including the start date, and periodicity.
 - **View Habits:** Displays all habits by name, start date, and periodicity.
 - **Analyze Habit:** Provides analysis options for a specific habit, the longest run streak for all habits, or for a given habit, alternatively to list habits by periodicity.
 - **Log Entry:** Enables the user to log a completion date and time for a habit.
 - **Delete Habit:** Permits users to remove habits they no longer want to track.
 - **Exit:** Users can exit the application at any time.

Tools and Libraries

The application will be built in PyCharm IDE using Python 3.7 or later. The following tools and libraries will be used:

- **JSON:** For data storage and retrieval.
- **questionary:** For creating interactive CLI prompts.
- **pytest:** For unit testing the application.
- **requirements.txt:** To manage and install the necessary dependencies and libraries.

Documentation

The application will be documented in a README.md file, a Markdown file compatible with GitHub, offering step-by-step instructions on installation, usage, and troubleshooting. The project will be uploaded from PyCharm to GitHub at a later stage of the development.

Testing

Making sure all key features of the habit tracking app are thoroughly tested will provide the confidence that the code works as expected.

- Creation of habits.
- Calculation of streaks (daily and weekly).
- Logging entries (daily and weekly with date and time stamp).
- Saving to and loading from the database.
- Adding and deleting habits.
- Getting habits by periodicity.
- Getting the longest streak for all habits and for a specific habit.

Conclusion

The planning phase has laid out the main parts, how data will be stored, how users will interact with the app, and the analytical features needed for the habit tracking app. The flow chart shows how users will move through the app and interact with it. By thoroughly planning and designing these elements, a good base is created for the development stages. This preparation should help avoid problems and improve the app's overall quality and functionality.